

TrES-2b

R-0070

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_210409 · created 2026-07-15T16:31:30+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459313.92665 +/- 0.00076 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1485 +/- 0.0079
Transit depth (Rp/Rs)^2	2.2 +/- 0.24 [%]
Orbital Inclination (inc)	83.95 +/- 0.36
Airmass coefficient 1 (a1)	1.0628 +/- 0.0082
Airmass coefficient 2 (a2)	-0.0438 +/- 0.0054
Scatter in the residuals of the lightcurve fit is	0.78 %
Best Comparison Star	#2 - [40, 147]
Optimal Method	PSF photometry
Transit Duration (day)	0.0786 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1485 ± 0.0079 vs 0.1254 (deviation 2.9σ)
Duration fit vs published	0.0786 d vs 0.0746 d (deviation 0.71σ)
Mid-time O–C	1.9 min
Depth SNR	18.7
Residual scatter	0.78 %

Light curve

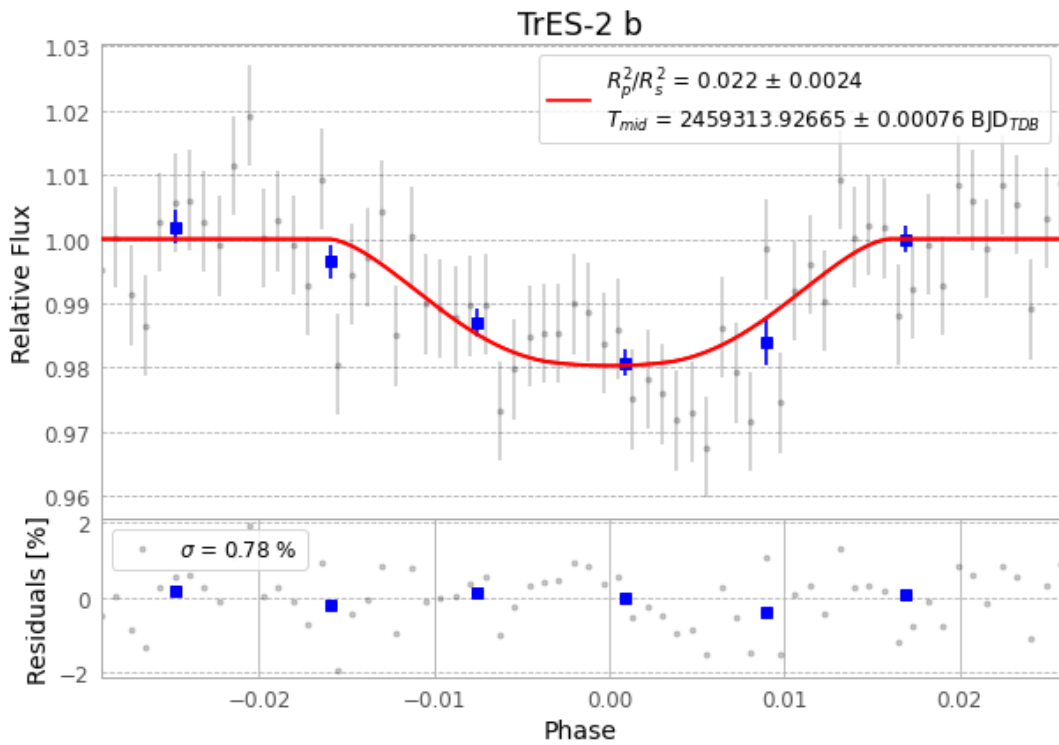


Figure 1 — FinalLightCurve_TrES-2 b_2021-04-09.png (SHA-256 8adcdcbd471d0210...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 281], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b4c0a8b249a9fba7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

66 FITS frames (43 MB), TRES-2210409083012.FITS → TRES-2210409114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-210409.log (8ad01de285a8...), FinalLightCurve_TrES-2 b_2021-04-09.png (8adcdcbd471d...), FinalLightCurve_TrES-2 b_2021-04-09.csv (a4b223b59c6e...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.